

How Much Damage Can GenAI Repair? Fact Recovery in Damaged Historical Newspaper Text

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Abstract

[TODO: Write abstract after results are final.]

1 Introduction

Digitised newspapers are now a core source for historical research. Digitisation lets scholars search and mine large collections by keyword rather than reading page by page (Nguyen et al., 2021). Yet what historians search and mine is not the page itself but the text that an OCR engine read from a scan of it. OCR errors measurably change the results of quantitative analyses (Hill and Hengchen, 2019) and harm downstream NLP tasks (Strien et al., 2020), with named entities among the most affected (Hamdi et al., 2023). In BLN600, a corpus of 19th-century British newspaper excerpts, the mean character error rate (CER) is 7%, but between excerpts of a single title it ranges from 0.3% to 44.5%, and pages too faded or damaged to read had to be left out of the corpus altogether (Booth et al., 2024). For the British Library’s own 19th-century newspaper archive, character accuracy reaches 83.6% but word accuracy falls to 78%, and accuracy for capitalised words and number groups drops further still, to 63.4% and 64.1% (Tanner et al., 2009). The words hit hardest are thus often the facts a historian searches for: names, places, dates and sums of money.

Generative AI makes it tempting to repair such text automatically, but the evidence is mixed. Llama 2, instruction-tuned on BLN600, reduced the CER of its OCR by 54.5%, against 23.3% for a fine-tuned BART (Thomas et al., 2024), while a study of 14 LLMs on several historical benchmarks found them “anything but efficient” at post-correction: they usually degraded the input, often by inventing text (Boros et al., 2024). This points to a deeper risk. A generative model always returns fluent text, even where the evidence for a word is gone (Ji et al., 2023). For an archive, a plausible

but wrong name is worse than visible noise, because the reader can no longer see that anything was damaged. Similarity-based metrics may hide this error too: BERTScore, for example, rates a translation that says *Tuesday* where the reference says *Monday* as highly similar (Zhang et al., 2020).

Existing post-OCR studies report error rates over whatever noise their OCR happened to produce. Where they break results down by noise level, that level is observed rather than controlled, and facts are not scored apart from the rest of the text. They thus show that a model helps on average, but not how far repair can be trusted as damage grows, nor whether the facts inside the damaged region survive. We therefore ask:

RQ. When the damaged words of a sentence are known and their garbled letters are visible, how much of the original wording, and in particular its fact words such as names, places and numbers, can a fine-tuned masked language model and a few-shot instruction-following LLM restore, and how does this change as the damaged share of the sentence grows?

We split it into three sub-questions. **SQ1** (severity): does repair quality fall smoothly as the damaged share of a sentence grows, or break down beyond some level? **SQ2** (method): given identical input, does a fine-tuned masked language model or a few-shot LLM repair better? **SQ3** (metric validity): does semantic similarity overstate repair quality compared with exact fact recovery? We fixed four hypotheses after tuning on dev but before computing any test score. **H1**: repair quality falls as the damaged share grows, and Fact Recovery Rate falls faster than BERTScore. **H2**: with identical input, the few-shot LLM restores more fact words than fine-tuned BERT, because it reads all garbled forms of a sentence together. **H2b**: GenAI repair beats a plain dictionary lookup on fact words, not only on

common words. **H3:** BERTScore overstates repair quality: at every level, some repairs with a wrong fact score as high as repairs with the right one.

Answering these questions needs damage whose amount we control and whose original text we know, which real damaged pages cannot give. We therefore build a controlled experiment on BLN600 (Booth et al., 2024): in each of 150 test sentences we damage one block of words around a fact at five nested levels, from one word to 75% of the sentence, with character noise calibrated on the corpus’s own OCR errors. All methods see the same input, the intact context plus the garbled form of each damaged word, so only the repair differs. We compare masked infilling with BERT (Devlin et al., 2019), fine-tuned on in-domain gold text and guided by letter similarity and dictionary candidates in the spirit of “context beats confusion” (Evershed and Fitch, 2014), with an open-weight LLM prompted with five solved examples. As reference lines we add a dictionary lookup that uses no model at all and, descriptively, BART (?) fine-tuned as a denoiser. We score the repaired words with BERTScore (Zhang et al., 2020) for meaning and with a Fact Recovery Rate for exact facts, and probe the LLM for memorised test text (Sainz et al., 2023). Repair works on text only and assumes the damaged region is known; detecting it, and repairing from the scanned image with a vision-capable model, are left to future work.

Our contributions are (i) a fact-centred, graded damage benchmark on BLN600 with nested spans, so the same place gets worse from level to level; (ii) a controlled comparison of masked infilling and few-shot generation under identical input and scoring, measured against a non-GenAI lookup; and (iii) a fact-level metric next to BERTScore that shows where semantic similarity and factual accuracy part ways. [TODO: Key outcomes, from the TEST split only: (1) the level at which Fact Recovery Rate falls to the no-repair floor; (2) which method wins at low vs high damage, and whether a crossing is a format-failure effect; (3) the size of the BERTScore vs Fact Recovery gap.]

2 Related Work

SQ1: grading damage for repair. We know of no study that controls damage severity for the repair step itself; graded severity has so far been used only for *reading* OCR’d text. Post-OCR research splits the task into error *detection* and error *cor-*

rection (Nguyen et al., 2021), scored separately in the ICDAR 2019 competition (Rigaud et al., 2019); we study correction only and assume the damaged span is known. Results on how well GenAI corrects disagree. An instruction-tuned Llama 2 cut CER on BLN600 further than a fine-tuned BART (Thomas et al., 2024), while 14 prompted LLMs mostly degraded historical transcripts (Boros et al., 2024); on early modern Dutch, fine-tuned seq2seq models beat fine-tuned generative ones (Debaene et al., 2025); and prompted open-weight LLMs lowered CER for English but not for Finnish (Kanerva et al., 2025). These results diverge by training regime, language and source quality, but in all of them the amount of damage is whatever the OCR produced: it can be observed, not set, so they cannot show at what point repair stops working. Controlled severity levels do exist for reading OCR text: page images degraded at two levels per degradation type and then OCR’d have been used to measure named-entity recognition (Hamdi et al., 2020), and real OCR output has been binned into quality bands for other downstream tasks (Strien et al., 2020). SQ1 applies this grading logic to the repair step.

SQ2: restoring text at known gaps. The closest prior task is restoring ancient text at known positions. Pythia predicts 1–10 hidden characters in Greek inscriptions (Assael et al., 2019), Ithaca extends this and pairs predictions with historians’ judgement (Assael et al., 2022), and a masked language model fills gaps in Akkadian texts, taking the number of missing signs as given and extending token predictions to multi-sign gaps with a beam search, with accuracy falling as the gap grows (Lazar et al., 2021). These studies differ from ours in three ways. They hide characters completely, whereas OCR damage leaves letters garbled but partly readable, which our slot input exploits (Sec. 3.3); their gaps are at most a few tokens, while ours reach 75% of a sentence (SQ1); and none compares a masked language model with a prompted LLM (SQ2) or scores facts apart from overall similarity (SQ3). On the model side, BERT fills masked positions in parallel (Devlin et al., 2019), which makes multi-token blanks harder to fill (Shen et al., 2020); our search over candidate lengths (Sec. 3.4) follows the beam decoding of Lazar et al. (2021). Left-to-right infilling (Donahue et al., 2020) is the natural counterpart for a prompted LLM, and our reranking adopts the classic OCR idea of weighing context against letter

evidence (Evershed and Fitch, 2014). Filling a blank from context is generation conditioned on the rest of the sentence (Donahue et al., 2020), and BERT can be sampled to produce text (Wang and Cho, 2019), so both methods count as GenAI here.¹

SQ3: meaning versus facts. A similarity score can pass a repair that gets the facts wrong, and SQ3 tests whether that happens here. BERTScore (Zhang et al., 2020) is the standard similarity metric for generated text, but in summarisation it correlates only weakly with human judgements of faithfulness (Spearman 0.19) and factuality (0.12) (Maynez et al., 2020), and in machine translation it fails to penalise a wrong candidate that shares most content words with the reference (Hanna and Bojar, 2021), in part because BERT represents names and numbers poorly (as reviewed *ibid.*). A repaired span that restores every word but one name is exactly such a candidate. Since this evidence comes from other tasks, SQ3 checks it directly by scoring the same repairs with BERTScore and with the Fact Recovery Rate. A second concern is memorisation. Thomas et al. (2024) chose BLN600 over the ICDAR benchmarks to avoid contamination, but BLN600 itself has been public since 2024, so a newer LLM may have seen it in pretraining (Sainz et al., 2023). We probe for this with a simplified form of guided completion (Golchin and Surdeanu, 2024), prompting the model with the start of a test sentence and comparing its continuation with the true remainder.

No single element of our design is new; the combination is. Table 1 places our study against the three closest lines of work, and Section 3 builds each missing element: graded nested damage (Sec. 3.2), a shared slot input (Sec. 3.3), two repair families and a non-GenAI lookup under identical input (Sec. 3.4), and a fact-level score next to BERTScore (Sec. 4.2).

3 Method

3.1 Task and Data

Given a sentence whose damaged span is known and whose garbled letters are visible, the task is to predict one word per damaged position; the number of damaged words is therefore also given. All

¹We cite Wang and Cho (2019) only for this narrow claim; its Markov-random-field derivation was later shown to be incorrect by one of its authors. Read sceptically, BERT is the fine-tuned in-domain baseline the few-shot LLM must beat, which is how our experiment treats it either way.

Work	Known loc.	Letters visible	Graded severity	Fact score	MLM vs LLM
Post-OCR LLMs ^a	×	✓	×	×	×
Pythia ^b	✓	×	×	×	×
Akkadian MLM ^c	✓	×	✓	×	×
Ours	✓	✓	✓	✓	✓

Table 1: Position of this study. ^aThomas et al. (2024); Boros et al. (2024); Kanerva et al. (2025) (real OCR, error rates over the corpus). ^bAssael et al. (2019) (1–10 hidden characters). ^cLazar et al. (2021) (masked signs, results by gap length). Graded: amount of damage set by the experimenter and results reported per level.

repair methods take the same input and are scored by the same code (Sec. 3.3). We build this task on BLN600 (Booth et al., 2024), 600 excerpts of London crime reporting from British newspapers of 1834–1894, each with its source image, OCR output and a gold transcription. We split excerpts into sentences with pysbd and pool sentences of 20–60 words with at least two *fact tokens*: a number, any word containing a digit or pound sign, or a capitalised word that is not the first word of the sentence (a short stop list excludes titles such as Mr, Mrs, The). This is a surface rule, not a named-entity tagger: it catches “Old” in “Old Bailey”, misses a lower-cased fact, and flags dates only through capitalisation or digits. [TODO: Optional: manual precision on 100 sampled fact tokens, e.g. “X% are names, places, numbers or dates”.] The pool holds 3,123 sentences from 592 excerpts. From it we draw, with a fixed seed, 150 test and 20 dev sentences from 157 excerpts (at most three per excerpt, no excerpt in both splits, balanced across three sentence-length bands); dev is used only for tuning and is never scored as a result.

3.2 Damage Simulation

For each sentence we draw one anchor among its fact tokens – always damaged, never dropped – and damage one contiguous span of k words that centers on it, with k set so the span covers 1 word, then 10%, 25%, 50% or 75% of the sentence (half-up rounding). Every larger span centers on the same anchor and fully contains every smaller one (Figure 1), so a higher damage level adds more damaged words around a fixed center; it does not make any single word more damaged; the per-word alteration intensity below is drawn the same way regardless of the span size. Each damaged word may be dropped outright (except the anchor). A surviving word has 20–30% of its letters and digits

Damage	Type	Value
Word drop	probability	0.10
Letter deletion	probability	0.15 ^a
Letter substitution	probability	0.85
confusion table	share of subst.	85.7%
capitalization	share of subst.	3.6%
look-alike map	share of subst.	4.9%
random character	share of subst.	5.8%

Table 2: Damage operations. Word drop and letter deletion are sampled per word and per altered character (^adeletion capped at 2 per word). Letter substitution is split among four sources by a fixed lookup order (confusion table, then capitalization, then look-alike map, then random), tried until one applies – the “share of subst.” rows are realized frequencies of that lookup, not independent sampling probabilities. The confusion table is calibrated on 29 held-out BLN600 excerpts.

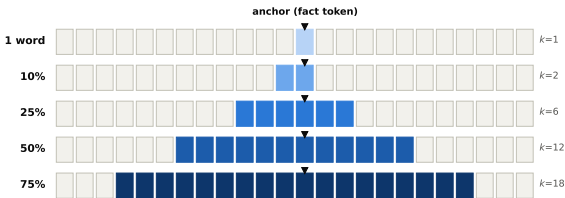


Figure 1: The damaged span at each level, for one example 24-word sentence. Every span centers on the same anchor word (the seeded fact token) and fully contains every smaller span, so a higher level does not change how heavily any single word is damaged – only how many words around the anchor are included.

altered, punctuation untouched, split between deletion and substitution as in Table 2 (the substitution source is a fixed lookup order – confusion table, then capitalization, then a look-alike map, then a random character – not a random draw between them). The 20–30% band was chosen to sit around the median per-word character error rate we measured on 29 held-out BLN600 excerpts (0.25, IQR 0.17–0.40). All calibration comes from real OCR errors on aligned words, so it models the kind of noise OCR output already contains.

Table 3 shows the result on the test split: sentence-level CER grows roughly in step with real BLN600 OCR error rates, but the share of damaged words that are themselves facts falls from 100% at the 1-word level to 17% at 75%. About one in ten fact slots is dropped outright, leaving no visible letters for that fact at all.

3.3 Shared Slot Input

All methods see the same *slot view*: the sentence’s undamaged context is left untouched, and each

damaged gold word is replaced by a slot showing its garbled form (a dropped word becomes an empty slot, written $\langle ? \rangle$). This view is built from the gold tokens and the per-word edit operations, never from the raw corrupted text, so it cannot leak an easier version of the task. A predicted word is spliced back inside the original word’s edge punctuation, so context words are never rewritten and every difference in score comes from what a method puts in the slots. For example, the damaged sentence “A POWERFULLY-BUILT man, $\langle \text{same} \rangle$ $\langle \text{Bbonias} \rangle$ $\langle \text{Huicl} \rangle$ $\langle \text{e} \rangle$ commission agent, . . .” has the gold continuation “named Thomas Hurd, a”; the garbled forms are visible to both repair methods, but the gold words are not. Showing garbled rather than blanked letters keeps this an OCR-repair task rather than blind text infilling. Treating the damaged region as known follows the usual split of post-OCR work into detection and correction (Nguyen et al., 2021; Rigaud et al., 2019) and the known-position setting of text restoration (Assael et al., 2019; Lazar et al., 2021); it isolates repair from detection, which we leave to future work.

3.4 Repair Methods

(a) **BERT + reranking.** We fine-tune bert-base-cased (Devlin et al., 2019) (110M parameters; cased because fact tokens are partly defined by capitalisation) on clean gold text from the 443 excerpts outside the sample (2,939 sentences; a further 10% of these excerpts held out for validation, with zero excerpt or sentence overlap with test or dev, which we verified directly). Training mirrors inference: for each target word, span words before it are left as gold text, the target is masked with its true number of subword pieces, and later span words each become one [MASK]. Fine-tuning settings are in Table 4; the best checkpoint (epoch 2) lowered the per-piece validation loss from 5.61 (pretrained) to 4.14, and training loss kept falling while validation loss rose again over the next two epochs, confirming the early stop. At inference, slots are filled left to right, following the beam decoding of Lazar et al. (2021) with the settings in Table 4. A slot with a visible internal hyphen or apostrophe is split into sub-slots filled the same way; only 0.9% of slots (3.5% of fact slots) still need more than three pieces. Each candidate is scored as its mean log-probability per piece plus $\lambda(1 - d)$, where d is the normalised Levenshtein distance to the garbled form, so a fluent completion is pulled back

toward what the damaged letters actually show. This is a direct, modern instance of “context beats confusion” (Evershed and Fitch, 2014). We chose $\lambda = 8$ from a grid over $\{1, 2, 4, 8, 16\}$ on dev, where it gave the best exact-word rate at a point where Fact Recovery Rate had already plateaued. Dropped slots, which have no garbled letters to match against, are scored on the language-model term alone. We keep this context-only variant as an ablation (*BERT, no dict.*).

(b) BERT + dictionary. Reranking cannot choose a word that BERT never proposes: on dev, BERT’s top candidates for *Wrighb* (gold *Wright*) were *Smith*, *White* and *andooll*. Our main MLM method therefore adds, for each slot, the five entries of a word list that are closest in spelling to the garbled form to BERT’s candidates, and scores them with the same formula. The list holds the 16,469 distinct words of the gold text of the 443 excerpts outside the sample, so no dev or test text enters it. On a dev grid over list size (5, 10, 20, 50) and λ (1 to 128) we chose 5 entries and $\lambda = 64$ by the same rule as above. At this weight the letters decide most slots, and BERT’s context breaks near ties and fills dropped slots.

(c) LLM few-shot. We prompt Qwen3-30B-A3B-Instruct-2507 (?), a mixture-of-experts model with 30B parameters of which 3.3B are active per token, served through GWDG’s SAIA API. The prompt holds a system message, a task instruction, five solved dev examples (one per level, held out from all scoring), and the same slot view as BERT with numbered slots, and asks for a JSON list of exactly k strings at temperature 0. We first used Llama-3.1-8B-Instruct (Grattafiori et al., 2024), but its dev format-failure rates (21% and 27% for the two prompt variants) were above the 15% limit we had set in advance for switching model. We use only open-weight models on academic infrastructure, so that the licensed BLN600 text (CC BY-NC-ND) is never sent to a commercial third-party service. The system message was necessary: without it, an early dev check found the model answering in Python or tool-call syntax in 15 of 15 cases, in line with the finding that stating the input type and output format helps LLM post-correction (Boros et al., 2024). An answer of the wrong length, not valid JSON, or cut off by the token limit counts as a *format failure* and is scored as if no repair had been made, with no retry and no manual correction. Because output formatting alone can swing LLM post-correction results (Kanerva et al., 2025), we

Level	1w	10%	25%	50%	75%
Sentence CER (%)	1.0	3.2	7.9	15.5	23.1
Facts in span (mean)	1.0	1.6	2.2	3.2	4.2
Facts / damaged words (%)	100	48	26	19	17

Table 3: Damage by level (test split). CER is over the whole sentence against gold; real BLN600 OCR is about 7% (Booth et al., 2024).

also report the LLM’s scores over valid answers only, which separates failing to repair from failing to follow the format. A secondary variant supplies the whole surrounding article as context instead of just the sentence; its other sentences are gold, not damaged, so it is an upper bound rather than a fair comparison with BERT, and we compare it only against the LLM’s own few-shot run. Zero-shot prompting was tried on dev and dropped, so this design cannot isolate how much of the LLM’s performance comes from the five examples rather than the instruction alone. The two methods also differ in supervision by design: BERT is fine-tuned on in-domain text, the LLM sees five examples. SQ2 therefore compares two practical repair set-ups, not two architectures under equal training.

(d) Dictionary lookup. To ask whether GenAI beats a spell checker, each garbled word is replaced by the closest entry of the same word list (Levenshtein distance, ties by frequency); a dropped slot stays empty. The lookup uses letters only and no context, the pure “confusion” side of Evershed and Fitch (2014). Its limit is coverage: on dev, the list holds 95% of damaged words but only 77% of damaged fact words.

(e) BART denoiser (descriptive). bart-base (140M) (?), the model family that Thomas et al. (2024) compared on BLN600, reads the slot view and writes the sentence back with one word per slot. We fine-tune it on the BERT training sentences, damaged by our own generator at all five levels with new damage in each epoch (best epoch 8 of 10). Unlike the other methods, BART is thus trained on exactly the noise process of the test data, so its scores are optimistic. It was added after the hypotheses were set, and we report it descriptively only.

(f) No repair. The damaged text itself is scored as if it were a method, giving the floor every repair has to beat. Table 4 summarises the settings.

<i>BERT (both variants)</i>	
Base model	bert-base-cased (110M), clean gold text
Fine-tuning	lr 5e-5, batch 16, early stop (patience 2, max 10 ep.), best epoch 2
Reranking	1-3 masks/slot, beam 5, top 10; $\lambda = 8$ (no dict.), $\lambda = 64 + 5$ list words (dict.)
<i>LLM few-shot</i>	
Model	Qwen3-30B-A3B-Instruct-2507 (SAIA)
Prompt	system msg + 5 dev examples, numbered slots, JSON
Decoding	temperature 0, max 1024 tokens
<i>BART and lookup</i>	
BART	bart-base (140M), same optimiser as BERT, trained on our damage; best epoch 8; greedy
Lookup	16,469-word list, Levenshtein, no context

Table 4: Implementation settings. λ and list size chosen on dev only.

Factor	Levels
Severity	1 word, 10, 25, 50, 75% of words; clean level 0 for the contamination probe only
Length band	20-29, 30-44, 45-60 words
Methods	BERT + dict.; LLM few-shot; dictionary lookup; no repair. Secondary: LLM + article, BERT (no dict.), BART (descriptive)
Data	150 test sentences, 750 rows per method; dev 20 (15 scored, tuning only)
Metrics	BERTScore, Fact Recovery Rate (primary); anchor recovery, exact match, pooled span CER, repair gain, format failures, LLM valid-only scores, sentence BERTScore, contamination (secondary)

Table 5: Experimental design.

4 Empirical Investigation

4.1 Experimental Design

The design crosses three factors inside the same 150 test sentences: five damage levels (1 word, then 10, 25, 50 and 75% of the sentence), three length bands (20-29, 30-44 and 45-60 words), and the repair methods of Sec. 3.4 (Table 5). A clean level 0 serves only the contamination probe (Sec. 4.2). All damage levels reuse the same sentence and the same anchor word, so every comparison across levels is paired: for one sentence we can ask directly whether repair got better, worse or stayed the same as damage grew, rather than comparing averages over different sentences. The 20 dev sentences, disjoint from test, were used only to pick settings (Sec. 3.4); five of them are the LLM’s examples, so 15 are scored. Each method’s setting was frozen by a code gate before its test rows were touched,

and all test scores were computed once, by one notebook, after every method’s predictions were stored.

Statistics. Slots of one sentence share their context, so the sentence, not the slot, is our unit. Every cell gets a 95% bootstrap interval over sentences (10,000 resamples, the same resamples for all methods; ??). We test only the two comparisons that a hypothesis named before the test run: BERT + dict. vs LLM (H2) and LLM vs lookup (H2b). Per level we use an exact McNemar test on anchor recovery (?) and a Wilcoxon signed-rank test on span BERTScore, with Holm’s correction (?) over the ten tests of each family. All other comparisons are descriptive.

4.2 Metrics

We score the repaired span, not the whole sentence, since only the span was touched. BERTScore F1 (Zhang et al., 2020) compares the repaired span with the gold span using roberta-large (layer 17), rescaled against a baseline so that a score can go below 0; this is our primary meaning-level metric. We also report sentence-level BERTScore, where the intact context can raise the score. Fact Recovery Rate is our primary fact-level metric: the share of gold fact tokens in the span that the repair restores exactly in their slot, punctuation ignored. Because the share of facts among damaged words falls as the span grows (Table 3), we also report recovery of the anchor alone, which is the same word at every level and so gives a clean paired series for SQ1, and we split Fact Recovery Rate into fact slots with visible letters and dropped slots, which the methods must fill from context alone.

As secondary metrics we report exact-word match (the share of all span words restored exactly), span CER pooled per level (all character edits divided by all gold characters of that level), repair gain (pooled CER before minus after repair),² the LLM’s format-failure rate and its scores over valid answers only (Sec. 3.4), and a contamination probe. The probe is a simplified guided completion (Golchin and Surdeanu, 2024; Sainz et al., 2023): the LLM receives the first half of a clean test sentence and is asked to continue it, and a continuation whose normalised Levenshtein similarity to the true remainder (cut to the same length) is at least 0.9

²We pool CER per level rather than averaging it per row, because on a one-word span a single wrong or over-long word can already give a row CER of 1 or more, so short spans would dominate the mean.

Level	BERT+dict.	LLM	Lookup	BART	None
<i>BERTScore (span)</i>					
1w	—	—	—	—	—
10%	—	—	—	—	—
25%	—	—	—	—	—
50%	—	—	—	—	—
75%	—	—	—	—	—
<i>Fact Recovery Rate</i>					
1w	—	—	—	—	—
10%	—	—	—	—	—
25%	—	—	—	—	—
50%	—	—	—	—	—
75%	—	—	—	—	—

Table 6: Test results, 150 sentences per level. None = no repair; BART is descriptive only. BERT without dictionary and LLM + article are reported in the text. [TODO: 95% CIs per cell (notebook 4); LLM and BART format failures per level; LLM valid-only scores (footnote).]

counts as near-verbatim. Unlike Golchin and Surdeanu (2024), the prompt does not name BLN600 and there is no second condition to compare with, so a formulaic phrase can pass the threshold by chance; we therefore read a low rate as reassuring and inspect any hit by hand. Because of limited API resources, the probe covers 116 of the 150 test sentences.

BERTScore and Fact Recovery Rate are reported side by side because they can disagree by design (SQ3): a repair that substitutes a wrong but plausible name can still score well on meaning while scoring zero on Fact Recovery Rate. To put one number on this gap (H3) without a threshold, we take for each method and level the chance that a repair with a wrong anchor gets at least as high a span BERTScore as a repair with the right anchor (ties count half), which is $1 - \text{AUC}$. A value of 0 means BERTScore always ranks the right fact higher; 0.5 means it cannot tell right from wrong facts.

4.3 Results

[TODO: Fill after the LLM test run and notebook 04 on test.]

5 Interpretation and Conclusion

[TODO: P1, after results.] The objective was to find how far text repair can be trusted as the damaged share of a sentence grows. [VARIANT A, breakpoint:] Repair holds up to [LEVEL]: there, the anchor fact is restored in [X]% of sentences by the better method, against [Y]% without repair. Beyond [LEVEL], anchor recovery falls to [Z]%,

close to the no-repair floor. [VARIANT B, smooth decline:] Anchor recovery falls steadily from [X]% at one damaged word to [Z]% at 75%, with no single level at which repair stops working. The objective is therefore reached gradually rather than completely: [one sentence on which part of the damage range repair can be trusted with]. [TODO: One line per hypothesis, from the test run: H1, H2, H2b, H3 supported, partly supported or not.]

[TODO: P2, first sentence after results: which method wins, at which levels, and whether a crossing survives on valid-only LLM scores.] Three differences between the methods can explain this pattern. First, the methods use the garbled letters differently: BERT never sees them as input, only through its candidate list and reranking term and one slot at a time (Sec. 3.4), and the lookup uses them alone, without context, whereas the LLM and BART read all garbled forms at once and can use one to interpret another. This should favour the LLM as spans grow. Second, BERT fills masked pieces in parallel, which makes multi-piece words hard to fill (Shen et al., 2020), and our search stops at three pieces per slot, so the 3.5% of fact slots that need more cannot be restored at all. Third, the methods differ in supervision and scale: BERT is fine-tuned on clean in-domain text and BART on our own damage, while the LLM has about 270 times more parameters than BERT (30 times more active per token) but sees only five examples. If BART leads, its training on the test noise process is the first explanation to rule out, not a sign of a better architecture. [If the contamination probe found hits: add that memorised text may inflate the LLM’s scores on the flagged sentences.]

[TODO: P3, after results.] [VARIANT A, gap found:] Span BERTScore stays at [X] at the [LEVEL] level while fact recovery drops to [Y]%, so semantic similarity overstates how much factual content is restored. For example, [one repair from the test split where a wrong but plausible name scores high on BERTScore and zero on fact recovery]. This matches earlier findings from summarisation and translation (Maynez et al., 2020; Hanna and Bojar, 2021) and shows that they carry over to text repair. [VARIANT B, no gap:] Span BERTScore and fact recovery fall together, so on repaired spans semantic similarity does not overstate factual accuracy. A likely reason is that span-level scoring leaves out the context that inflates sentence-level similarity (Sec. 4.2).

[TODO: P4, first sentence after results: the prac-

tical threshold, e.g. “names and numbers restored beyond [LEVEL] should not be trusted without a check”.] This threshold holds under our conditions: character noise calibrated on real but mild OCR errors, a damaged region that is known in advance, and newspaper pages legible enough to have been transcribed (Booth et al., 2024). On real damaged pages, the threshold is more likely to be lower than higher. Whatever the method, an archive should not replace damaged text silently. Repaired words should stay marked, so that a reader can still see where the evidence was damaged; Ithaca’s pairing of model predictions with historians’ judgement shows one way to do this (Assael et al., 2022).

[TODO: P5, one-sentence summary of the answer after results.] Three extensions follow directly from our assumptions. First, repair should be combined with detecting the damaged region, which we took as given. Second, the results should be checked on real OCR errors from damaged pages rather than on calibrated synthetic noise. Third, repair could draw on the scanned image itself, whether through image-level degradation as a test bed (Hamdi et al., 2020) or through vision-capable models that read the damaged page directly (cf. Kanerva et al., 2025).

Limitations

Damage. Our damage is synthetic character noise in one contiguous span, not a model of torn, stained or faded pages: intensity is drawn per word rather than spatially correlated as in a real torn patch, and the confusion table comes from mild OCR errors on aligned words, so it shows which characters get confused but likely understates how heavy real damage is. Synthetic and natural noise can differ enough that results on one do not carry over to the other (Belinkov and Bisk, 2017); degrading page images before OCR (Hamdi et al., 2020) would be a closer test. BLN600 itself excludes pages too faded or damaged to transcribe (Booth et al., 2024). *Task.* The damaged region and the number of damaged words are given, so no detection step is tested. Damage is centred on a fact by design, so facts make up 100% of damaged words at the 1-word level but only 17% at 75%, and what else falls in the span, a stock phrase or a rare name, is not controlled. *Measurement.* Fact tokens come from a surface rule and are scored by exact match, so a correct spelling variant (*M’Donald* for *McDonald*) counts as a failure. With 150 test sentences,

only 26 in the longest band, and one noise draw per sentence, we report trends rather than powered effects. *Comparison.* The methods differ beyond the repair step: the LLM and BART read the garbled letters directly, BERT only through its candidates and reranking score; BERT predicts a word’s pieces in parallel (Shen et al., 2020) and cannot produce words of more than three pieces; supervision, scale and pretraining differ; and the LLM may have seen BLN600, which has been public since 2024. Format failures count as no repair, and the article variant uses gold context and is therefore an upper bound. BART is trained on the same damage generator as the test data, so its scores are optimistic, and each fine-tuned model comes from one training run with one seed. The hypotheses were written after tuning on dev, though before any test score, and the contamination probe covers only 116 of the 150 test sentences. Finally, all data are English London crime reporting, mostly from two weeklies (572 of 600 excerpts), and we test a single LLM.

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Appendix Outline (for the defence, not part of this report)

At the defence we would add: the full damage parameters and confusion table, with the first, heavier noise setting we dropped; BERT and BART training curves; the λ and word-list grids chosen on dev; the full prompts, the first prompt without a system message and the Llama-3.1-8B dev run that led to the model switch; the dropped zero-shot run; results by length band and the full paired-test table; the manual check of 18 repairs; the probe details; and two ablations we planned but did not run (letters hidden, confusion-weighted reranking).